Geog 4/6300: Lab 3-Confidence intervals

Confidence intervals

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Overview:

In this lab, we will create confidence intervals for January precipitation in Washington during two time periods: 1950-1969 and 2005-2024. Using county-level precipitation data from the ERA5 dataset, you will estimate statewide precipitation patterns, then examine variation and change at the county level. We've summed the precipitation in January for each county over two twenty year time periods: 1950-1969 and 2005-2024. You'll be using those data to create confidence intervals at a state and county level.

To load these data, load the csv file, which is included in the data folder:

```
era5_data<-read_csv("data/precip_uswest_era5.csv")
```

This is a long format dataset including county identifiers, county population, median household income, and poverty rate. Census figures are based on the 2018-22 ACS. The last two columns (year, prcp_m, time_range) give the year and total precipitation in meters from the ERA5 NetCDF file, and two time ranges (1950-1969 and 2005-2024).

The script used to create this dataframe is also provided in this repository ('lab3_dataprep'), along with rasters of the yearly mean total precipitation for Januaries in the two time periods of interest.

For this lab to be marked complete, it must demonstrate the following learning standards:

- 11. Create effective data visualizations across multiple chart types (line, scatter, histogram, Q-Q plot), applying appropriate aesthetic choices such as color, grouping, and labeling. (Tasks 2 and 5)
- 12. Produce and interpret confidence interval plots that enable visual comparison of means and uncertainty across groups or time periods. (Tasks 2, 3, 5, and 6)

- 13. Visualize spatial data using appropriate mapping tools, selecting suitable classification schemes and cartographic elements to support clear communication. (Task 7)
- 14. Arrange and present tabular results in a readable format appropriate for a technical document or report. (Tasks 1 and 4)
- 15. Use measures of central tendency and dispersion to describe the variables in tabular datasets (Tasks 1 and 4)
- 20. Explaining how evidence from a sample supports conclusions about a population while acknowledging sampling variability and uncertainty. (Tasks 3 and 6)
- 22. Construct and interpret confidence intervals, explaining how factors such as sample size and spatial scale affect the precision of an estimate. (Tasks 1, 3, 4 and 6)

Part 1: Calculating statewide confidence intervals for Washington's precipitation

In the first part of this lab, you'll determine a confidence interval for Washington's precipitation for these two time periods. This can be helpful in identifying whether precipitation has changed across the state (e.g., whether the two confidence intervals overlap) using the county data as your sample.

Task 1 *Load the data and filter it to just include data for the state of Washington. Then create a table showing mean precipitation, standard deviation, number of observations (sample size), and upper/lower confidence interval bounds for each of the two time periods in the* `time_range` *field. Use* `kable` *to display the table when you are done.*

```
era5_data_Washington<-era5_data |>
  filter(st_name == "Washington")

era5_data_Washington_summary<-era5_data_Washington |>
  group_by(time_range) |>
  summarise(prcp_mean=mean(prcp_m),
            prcp_sd=sd(prcp_m),
            observations=n(),
            se=prcp_sd/sqrt(observations))

z.score_upper=qnorm(0.975)
```

```
z.score_lower=qnorm(0.025)

era5_data_Washington_summary<-era5_data_Washington_summary |>
  mutate(error=z.score_upper*se,
         ci_high=prcp_mean+error,
         ci_low=prcp_mean-error)

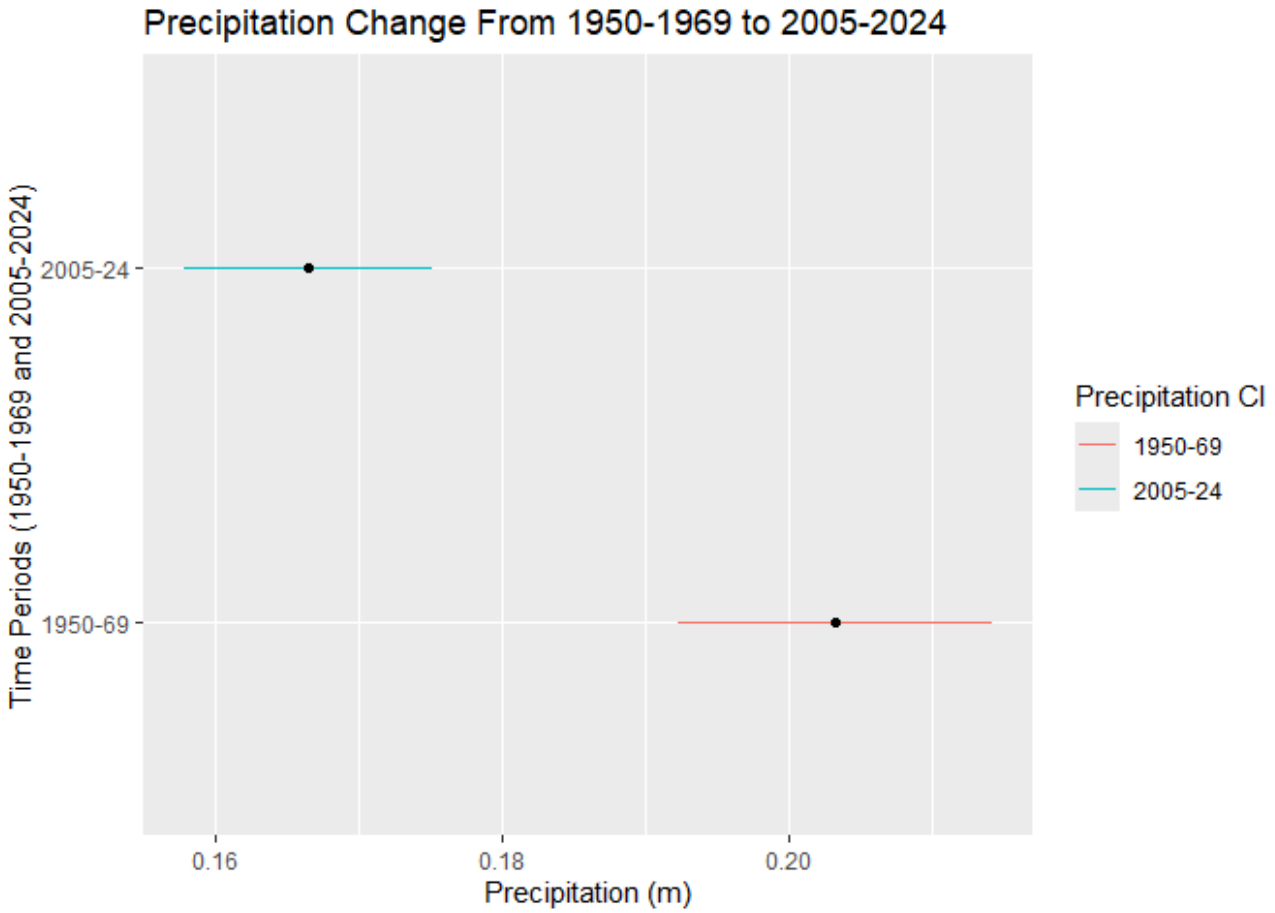
kable(era5_data_Washington_summary)
```

time_range	prcp_mean	prcp_sd	observations	se	error
1950-69	0.2032239	0.1562948	780	0.0055963	0.0109685
2005-24	0.1664626	0.1230138	780	0.0044046	0.0086329

Using the code provided in class, create a ggplot graphic that visualizes these two confidence intervals. You'll want to use `geom_point` and `geom_linerange` to do so. Use states as the y-axis and mean precipitation as the x axis. Vary the color and position of the dots and lines for the mean/CI by the `time_range` variable. Feel free to also experiment with axis labels/themes to make the graph more appealing and accessible.

Task 2 *Create a ggplot graphic as described above to show the confidence intervals for precipitation in Washington for each time range.*

```
ggplot(era5_data_Washington_summary) +
  geom_linerange(aes(xmin=ci_low, xmax=ci_high, y=time_range, group=time_range, color
  geom_point(aes(x=prcp_mean,y=time_range)) +
  labs(
    title = "Precipitation Change From 1950-1969 to 2005-2024",
    x = "Precipitation (m)",
    y = "Time Periods (1950-1969 and 2005-2024)",
    color = "Precipitation CI")
```



Task 3 *Interpret your table and graph. What do they tell us about the change in Washington's precipitation during this time?*

The table tells us that there was more precipitation on average in 1950 to 1969 than from 2005-2024, but there is a lot of variability in both time frames. The table and graph also show that the precipitation has changed significantly from the first time range (1950-1969) to the second (2005-2024). This is evident on the graph by the confidence intervals not having any overlap between them, meaning there is a significant change in the data. The data tells us the decrease in the mean precipitation from 1950-1969 to 2005-2024 is statistically significant, even though the standard deviation shows that the data itself is highly variable.

Part 2: Analyzing county data in Washington.

In part 1, you ran a state level analysis. What if you wanted to identify where populations might be negatively affected by precipitation changes within a state? In this section, you'll

look at the changes in precipitation **at county level** in Washington to

To start with, adapt your code from step 1 so that you create confidence intervals for **counties** instead of states. After creating that dataset, create another ggplot graph of those confidence intervals that you can compare to the state level data in Task 2 to examine how precipitation has changed across counties in Washington.

Task 4 *Create a summary table for January precipitation in Washington counties for both time periods. Use the same approach as in Task 1, but calculate statistics for counties rather than the state as a whole. Call the head of your table with* `kable` *when you are done.*

```
era5_data_Washington_summary_county<-era5_data_Washington |>
  group_by(time_range, cty_name) |>
  summarise(prcp_mean=mean(prcp_m),
            prcp_sd=sd(prcp_m),
            observations=n(),
            se=prcp_sd/sqrt(observations))
```

```
## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by time_range and cty_name.
## ℹ Output is grouped by time_range.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(time_range, cty_name))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
z.score_upper=qnorm(0.975)
z.score_lower=qnorm(0.025)

era5_data_Washington_summary_county<-era5_data_Washington_summary_county |>
  mutate(error=z.score_upper*se,
         ci_high=prcp_mean+error,
         ci_low=prcp_mean-error)

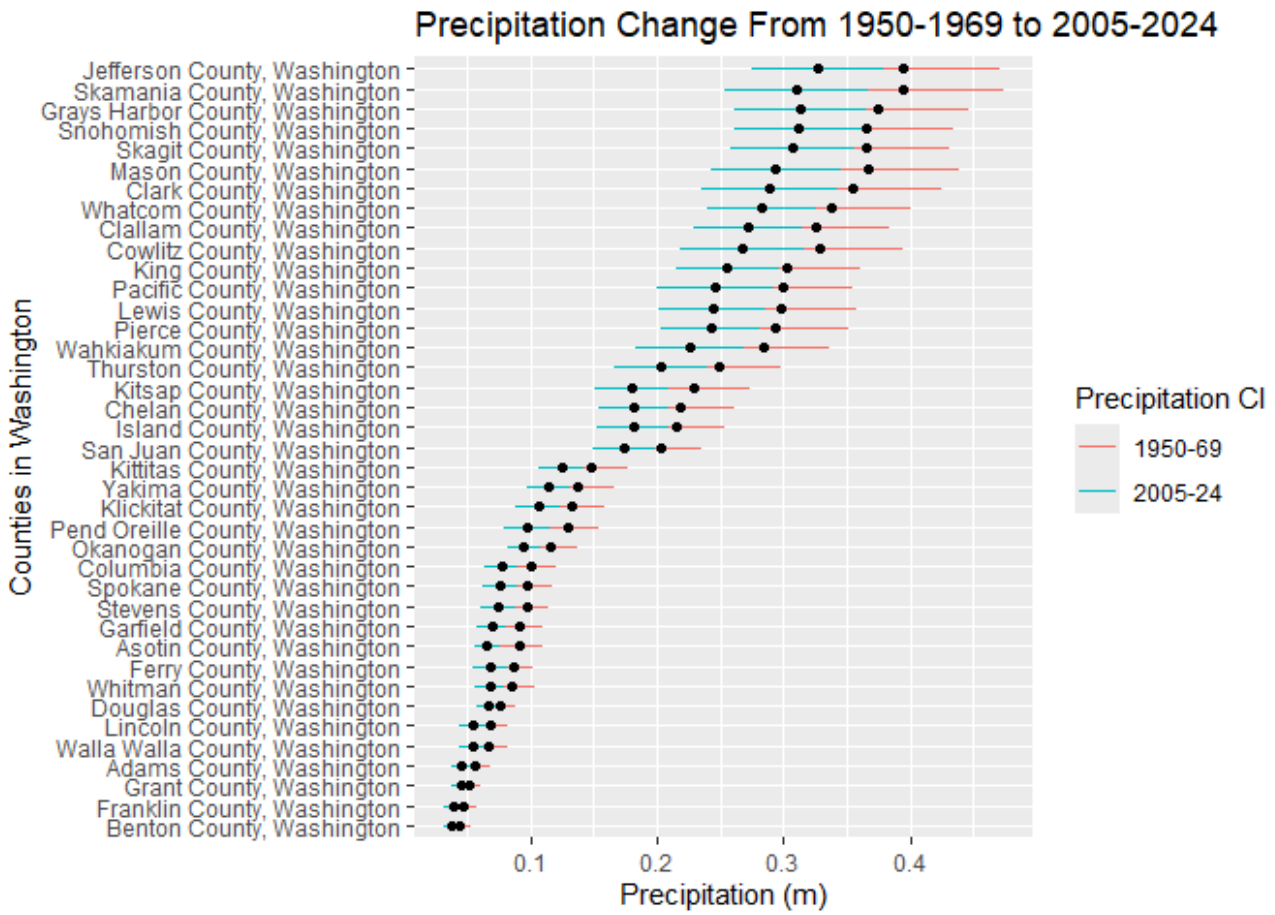
kable(head(era5_data_Washington_summary_county))
```

time_range	cty_name	prcp_mean	prcp_sd	observations	se
1950-69	Adams County, Washington	0.0561916	0.0275746	20	0.006165

time_range	cty_name	prcp_mean	prcp_sd	observations	se
1950-69	Asotin County, Washington	0.0917109	0.0412774	20	0.009229
1950-69	Benton County, Washington	0.0439175	0.0199657	20	0.004464
1950-69	Chelan County, Washington	0.2184237	0.0983827	20	0.021999
1950-69	Clallam County, Washington	0.3252100	0.1331571	20	0.029774
1950-69	Clark County, Washington	0.3545254	0.1621728	20	0.036262

Task 5 *Visualize the confidence intervals for the county-level data you generated in task 4, using the same style as you did for Task 2 above. For an optional extra challenge, visually highlight the counties where the confidence intervals between time periods do NOT overlap.*

```
ggplot(era5_data_Washington_summary_county) +
  geom_linerange(aes(xmin=ci_low, xmax=ci_high, y=reorder(cty_name, prcp_mean), group
  geom_point(aes(x=prcp_mean,y=cty_name)) +
  labs(
    title = "Precipitation Change From 1950-1969 to 2005-2024",
    x = "Precipitation (m)",
    y = "Counties in Washington",
    color = "Precipitation CI")
```



Task 6: *Compare the size of the error terms for your county level data to the error terms at state level that you calculated in Task 1. Are they bigger or smaller? Explain why the formula you used for calculating the error terms produced error terms of different sizes. How is this difference visible in your ggplot graphs?*

The error in the county level data is much smaller in general than the state level error.

The reason why the error for the county level data had generally larger errors than the state data is because it only had 20 observations per county, while the state data had almost 800 observations. The equation that causes this issue is the equation for the standard error (se=prcp_sd/sqrt(observations)). When you have a small number of observations, the denominator remains small and therefore the standard error stays large (dividing by 100 gives a smaller number than dividing by 10). Considering the confidence intervals is just the mean plus or minus the error, a larger error gives us a larger range for the confidence intervals, and therefore a larger visible error.

The difference is visible because the confidence intervals are larger and spread further apart (meaning the low limit of the confidence interval was not close to the upper limit of the confidence interval).

Task 7: *Transform the county data from task 4 into wide format so that the means for each time period are in the same row for each county. Then calculate the change in precipitation between the time periods. Join this table to the geopackage file that contains county boundaries and census data, making sure that you only have Washington counties included - use the geopackage* `data/acsdata_cty_westreg.gpkg` *. Then use* `tmap` *to map out the difference in precipitation means across the two time periods. Make sure the legend of this map is interpretable.*

```
library(terra)
```

```
## terra 1.9.46

##
## Attaching package: 'terra'

## The following object is masked from 'package:knitr':
##
##     spin

## The following object is masked from 'package:tidyr':
##
##     extract
```

```
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(tmap)

wa_cty <- st_read("data/acsdata_cty_westreg.gpkg") %>%
  st_transform(4326) %>%
  filter(st_name == "Washington")
```

```
## Reading layer `acsdata_cty_westreg' from data source
##   `C:\Users\wbm11\OneDrive\Documents\GitHub\geog4300_lab3\data\acsdata_cty_westreg
##   using driver `GPKG'
## Simple feature collection with 414 features and 8 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
```

```
## Bounding box:  xmin: -124.7322 ymin: 31.33223 xmax: -102.0416 ymax: 49.00237
## Geodetic CRS:  WGS 84
```

```
era5_data_Washington_summary_county_wide<-era5_data_Washington_summary_county |>
  select(time_range, cty_name, prcp_mean) |>
  pivot_wider(names_from=time_range, values_from=prcp_mean)

era5_data_Washington_summary_county_wide <- era5_data_Washington_summary_county_wide
  mutate(Change_cm = (era5_data_Washington_summary_county_wide$'2005-24' - era5_data_

era5_data_joined <- wa_cty |>
  left_join(era5_data_Washington_summary_county_wide, by = "cty_name")

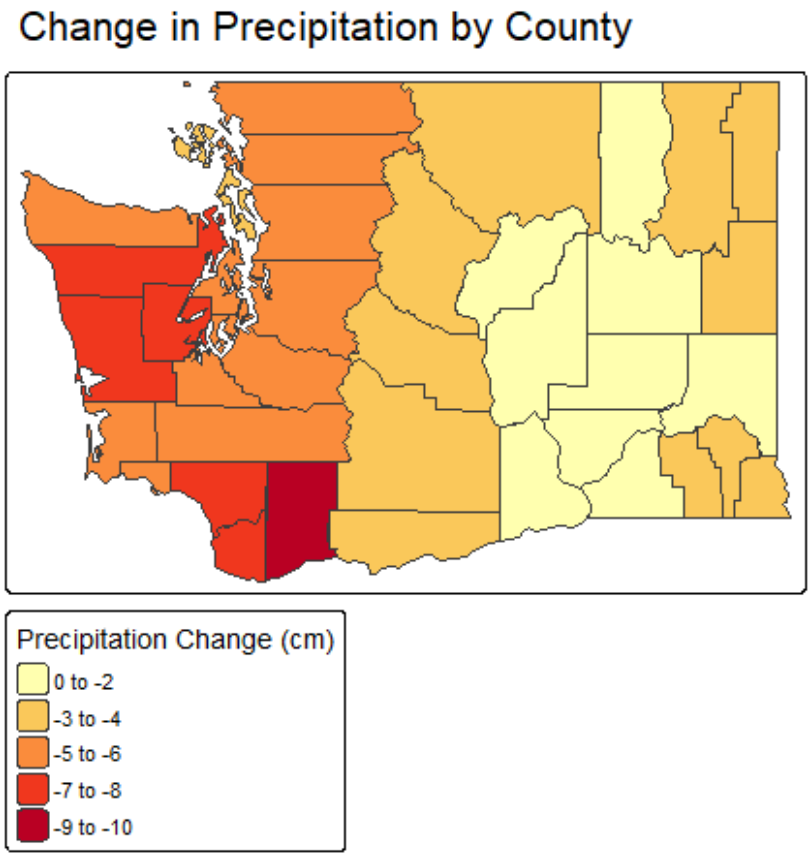
tm_shape(era5_data_joined)+
  tm_polygons("Change_cm", palette = "-YlOrRd",
              legend.format = list(text.separator = " to "),
              title = "Precipitation Change (cm)",
              textNA = FALSE,
              legend.reverse = TRUE,
              labels = c(
                "-9 to -10",
                "-7 to -8",
                "-5 to -6",
                "-3 to -4",
                "0 to -2")) +
  tm_layout(main.title = "Change in Precipitation by County")
```

```
##

## ── tmap v3 code detected ──────────────────────────────────────────────────

## [v3->v4] `tm_tm_polygons()`: migrate the argument(s) related to the scale of
## the map variable `fill` namely 'palette' (rename to 'values'), 'labels',
## 'textNA' (rename to 'label.na'), 'legend.format' (rename to 'label.format') to
## fill.scale = tm_scale(<HERE>).
## ℹ For small multiples, specify a 'tm_scale_' for each multiple, and put them in
##   a list: 'fill.scale = list(<scale1>, <scale2>, ...)'
## [v3->v4] `tm_polygons()`: migrate the argument(s) related to the legend of the
## map variable `fill` namely 'title', 'legend.reverse' (rename to 'reverse'),
## 'legend.format' (rename to 'format') to 'fill.legend = tm_legend(<HERE>)'
## [v3->v4] `tm_layout()`: use `tm_title()` instead of `tm_layout(main.title = )`
## Multiple palettes called "yl_or_rd" found: "brewer.yl_or_rd", "matplotlib.yl_or_rd
##
## [cols4all] color palettes: use palettes from the R package cols4all. Run
```

```
## `cols4all::c4a_gui()` to explore them. The old palette name "-YlOrRd" is named
## "yl_or_rd" (in long format "brewer.yl_or_rd")
```



Optional challenge section: Interpreting the implications of precipitation changes

Now that you've calculated these means and confidence intervals, let's identify who might be most affected by changes in precipitation. To see who is affected, create a dummy variable that flags any counties with a decline of 5 centimeters or more (0.05 meters) with a 1, labeling all other counties with a 0. Use group_by and summarise to do the following:

- Count the total population in each of these two county groups.
- Calculate the mean value for median income and poverty rate in these two groups.

Task 8 *Create a dummy variable for counties with a change of 0.05 meters or more between the time periods. Calculate how many people currently live in these counties along with the mean value of median income (medinc) and poverty rate (pov_pop_pct) for those areas.*

Task 9 *Create a one paragraph summary of your results from tasks 7 and 8. What do you learn about how precipitation has changed in the last 70 years across the state of Washington? How would you describe the population affected by large drops in precipitation based on your results?*

{Your response here.}

Small multiples or multi-panel maps are often a helpful way to show change over time. See this page for more information about how these are created in `tmap`:

- https://r-tmap.github.io/tmap/articles/basics_facets

Using these tools, create a small multiples map of your precipitation data for counties in a state of your choice for all 40 years in these two time periods. This means you will have one figure with 40 individual maps, all of the same state in different years. These maps should allow easy comparison across years and should be clearly labeled so that they could be interpreted by someone with no knowledge of this assignment.

In addition, describe two interesting trends you see in the figure you created.

{Your response here.}

Final submission stuff

Disclosure of assistance: Besides class materials, what other sources of assistance did you use while completing this lab? These can include input from classmates, relevant material identified through web searches (e.g., Stack Overflow), or assistance from ChatGPT or other AI tools. How did these sources support your own learning in completing this lab?

I actually did not use AI or assistance that much outside of class work. I did use AI for task 2 because I wanted an explanation on the formatting of the geom_linerange() tool, specifically how the reorder function worked. I also used AI to help find the best and most efficient way to change the color scheme for task 2 and 5. I did have to use it a couple times because of stupid mistakes like missing a pipe or plus sign (will I ever not make those errors?). Lastly

for AI, I used it to help format the figure for task 7. Since we wanted the figure and legend to be as easy to interpret as possible, I used AI to help me. First, I used it to help find the best way to assign a color scheme, then I used it because I did not like how the legend said “-10 - -9” because it looked like I was subtracting values, so I got it to help me make it “-10 to -9”, and then eventually “-9 to -10” because going in reverse made less sense. I used AI to reverse the legend because I wanted smaller integers on top and larger ones on the bottom (even though they are negative, I thought this flowed better). I also used AI to get over errors because I could not figure out the order in which I should do task 7. I used it to help set a plan for how I should write my own code, and it guided me by helping me format which steps I should do first.

I did have help from classmates. Grayson and I worked together on the lab as we both wanted to bounce ideas back and forth for task 6 and 7. I had the idea to change the text format for the legend to say “# to #”, and he came up with the idea to convert the legend and figure data to centimeters to get whole integers as well as which color scheme was best (we liked how it matches the drought monitor). We also discussed why the formulas gave larger errors to county data. We did also generally discuss the best ways to go about task 7, because we both had errors we were trying to deal with.

Lab reflection: How do you feel about the work you did on this lab? Was it easy, moderate, or hard? What are the biggest things you learned by completing it?

Honestly, outside of task 7, I thought this lab was much easier than lab 2. It may be because I have a good background in statistics from my time at LSU and because I like math equations, but I did not struggle with this lab much at all. I did like learning how to run statistics in R, and I did use the refresher in my thesis results, so I had a lot of enjoyment covering this. Overall, I feel good about my work on this lab.